



OMCS6250

Computer Networks

Interdomain Routing

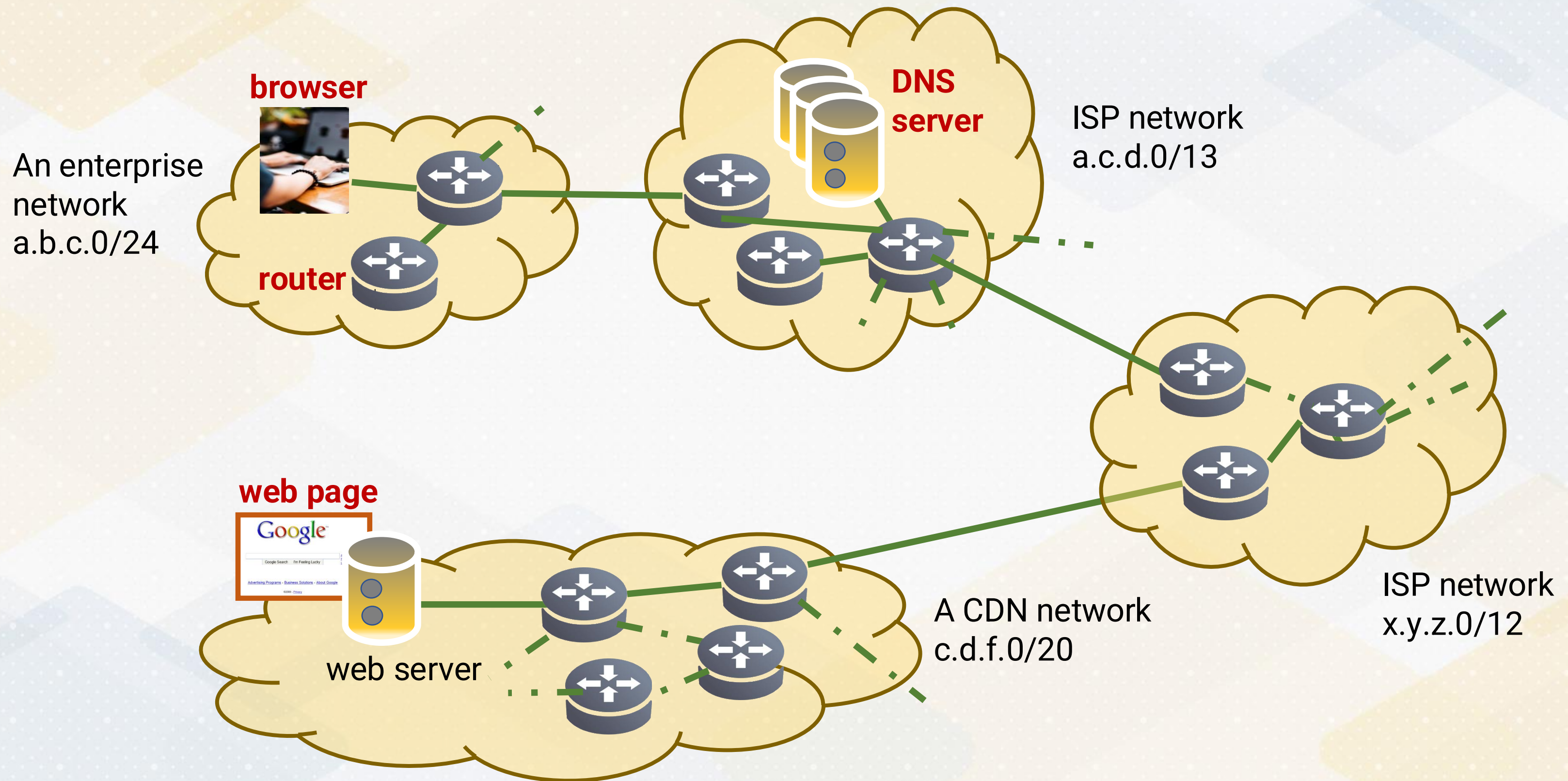
With Instructor Dr. Maria Konte

**BGP, Policies, AS Relationships, Path Selection
Process, IXPs, IXPs Route Servers**

Learning Objectives

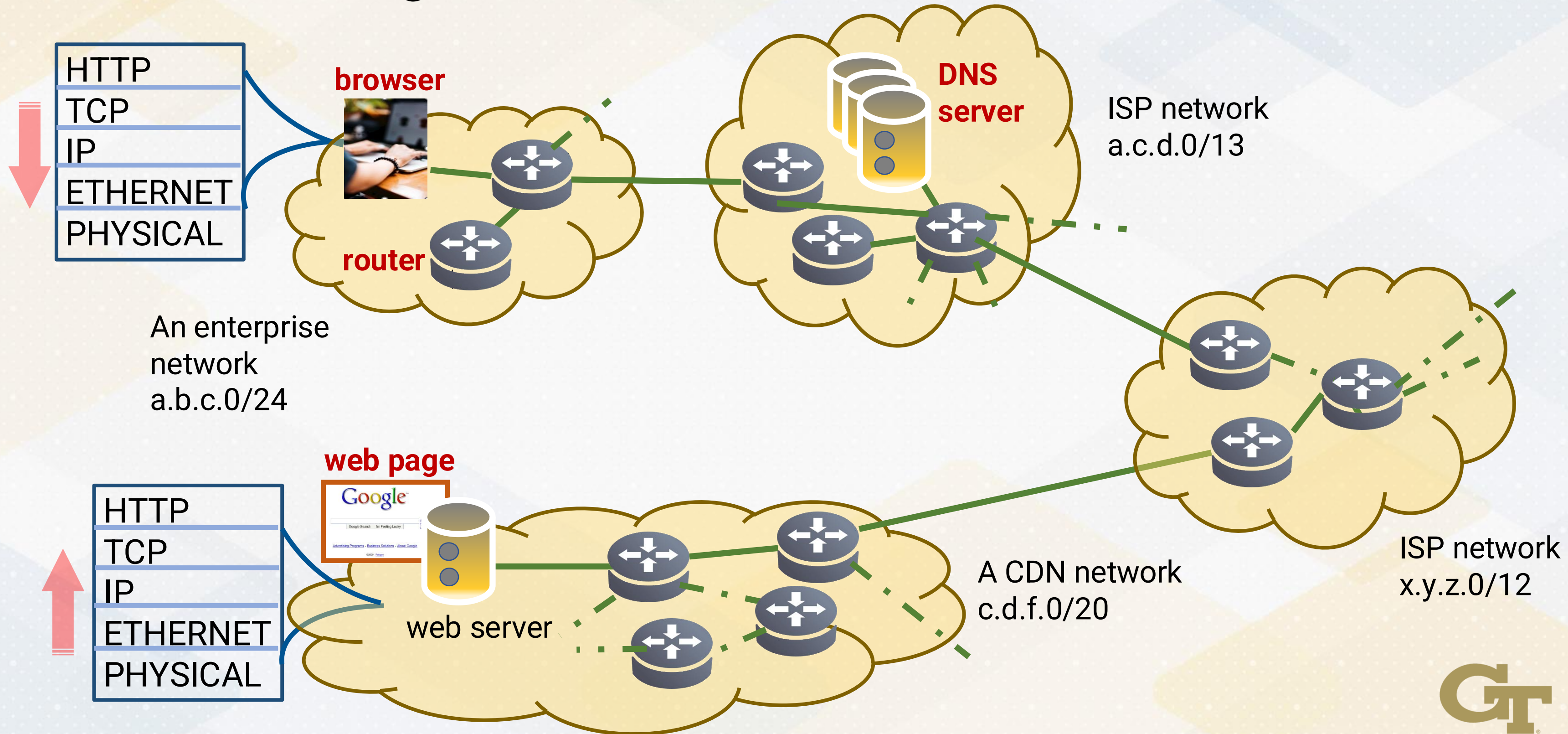
- Roadmap: **The Internet ecosystem** → AS relationships → policies → BGP → IXPs/route servers → failures
- Define AS, **intra vs inter routing** (IGP vs BGP)
- Explain **AS business relationships**: transit vs peering + incentives Apply import/export policy + route filters (customer/peer/provider)
- Describe **BGP sessions**, updates, path selection (Local Pref, MED)
- Explain why **IXPs** matter, how route servers scale peering
- Identify **operational risks** e.g. misconfigs

Reviewing the Internet's Protocol Stack



Reviewing the Internet's Protocol Stack

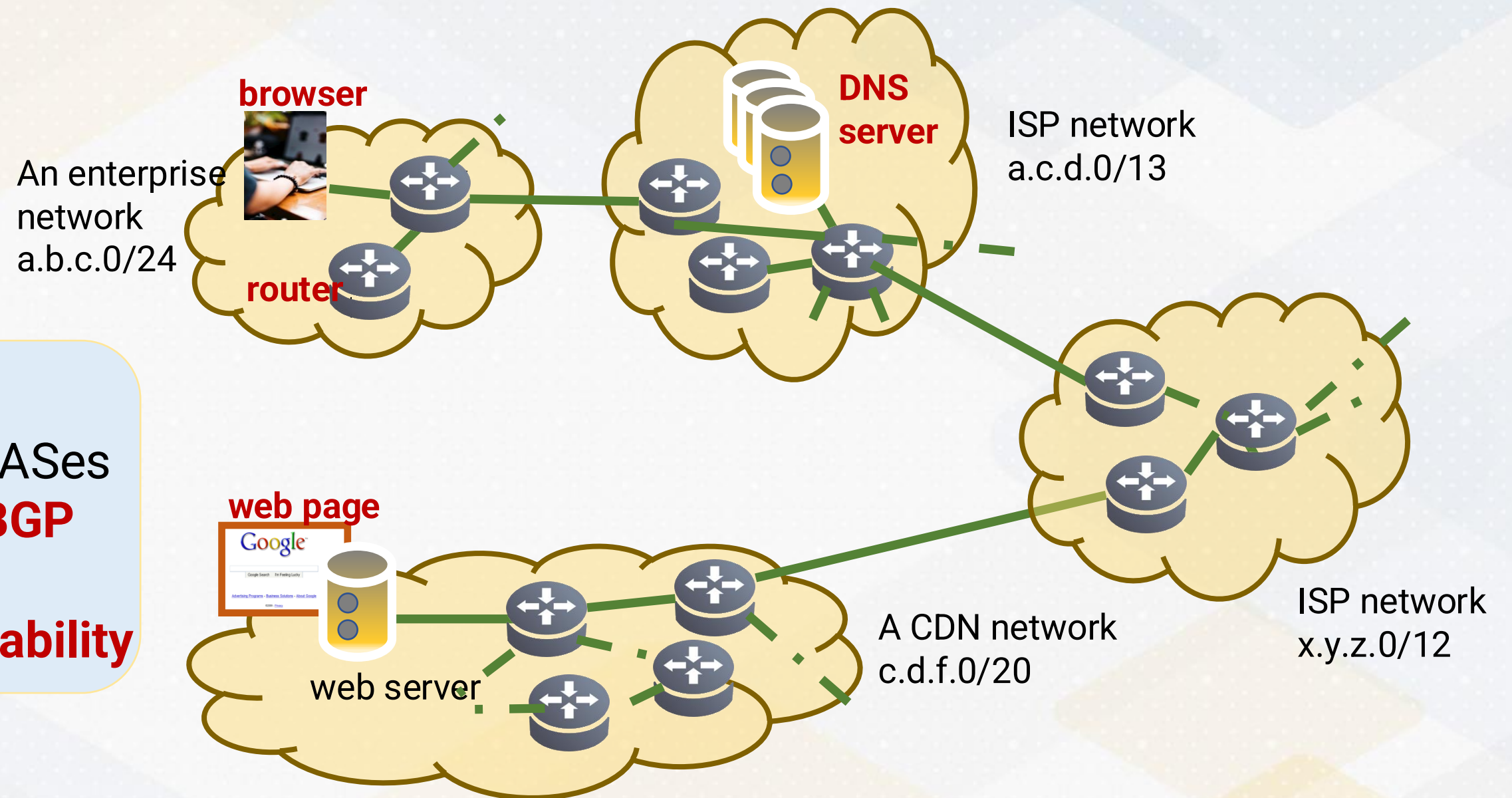
Interdomain Routing



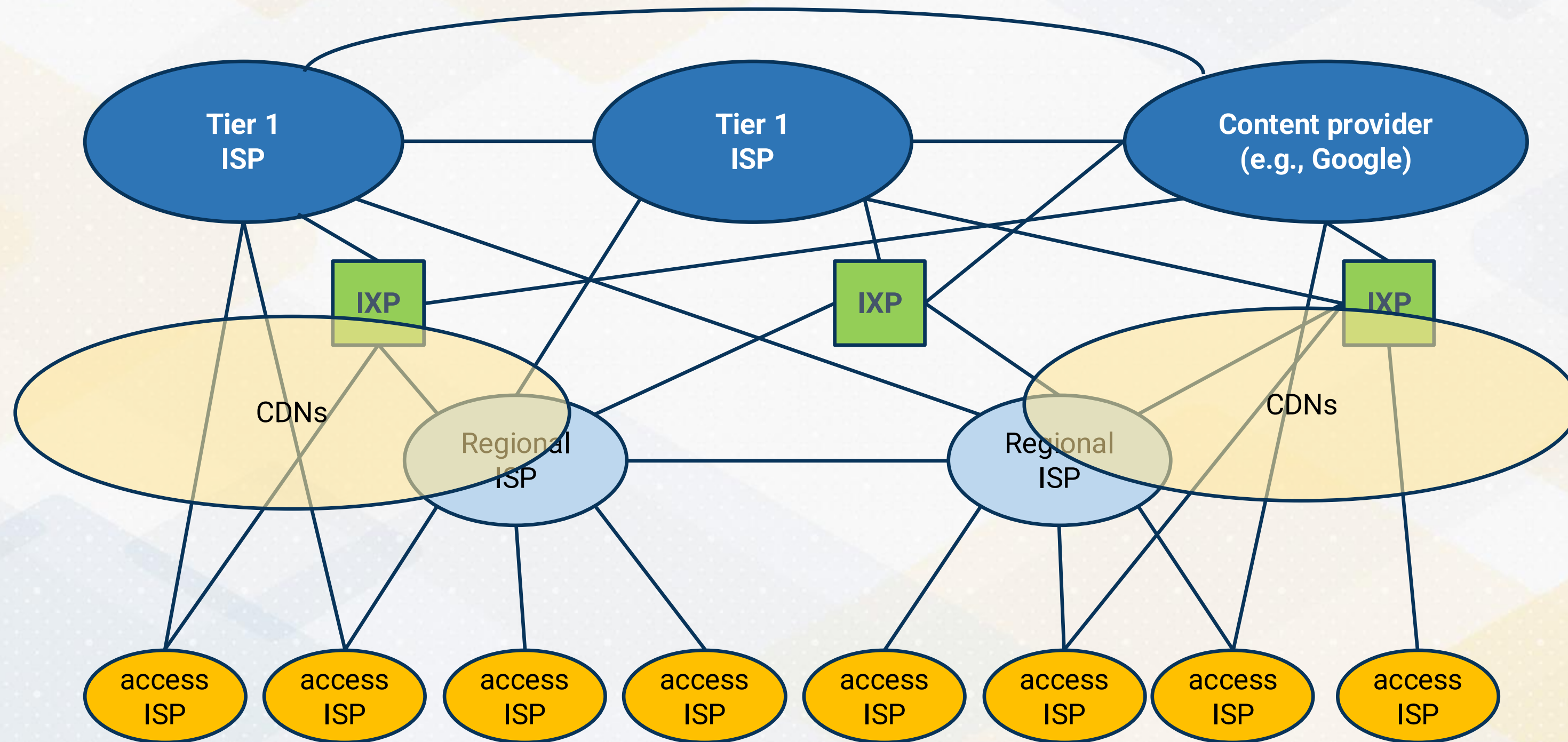
Reviewing the Internet's Protocol Stack

Interdomain Routing

- **Routing** is a network-layer (IP) function
- Two scopes: **within an AS** and between ASes
- IGP inside ASes (e.g., OSPF/IS-IS), and **BGP between ASes**
- Goal of interdomain routing: **policy, scalability**

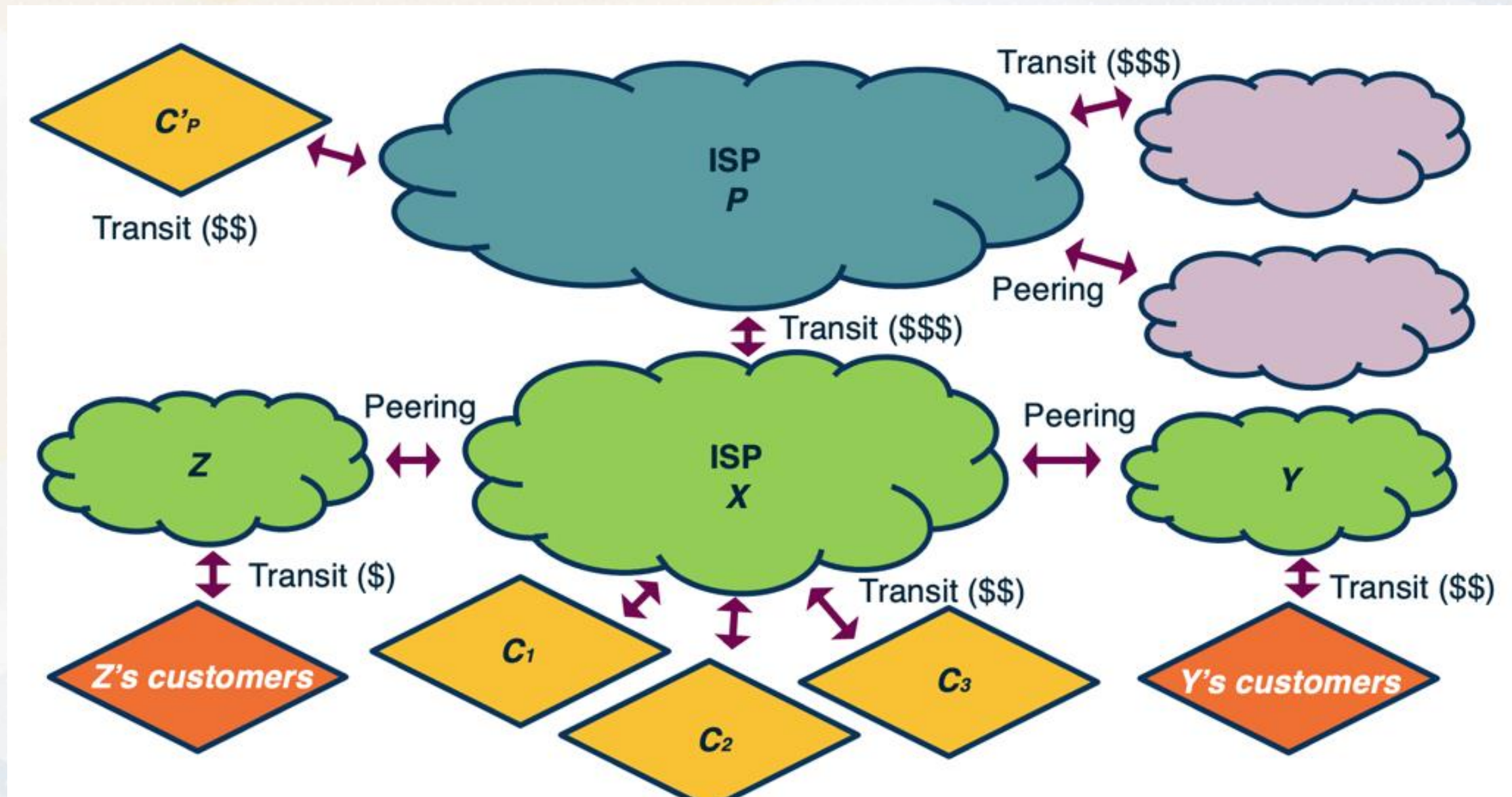


The Internet As An Ecosystem



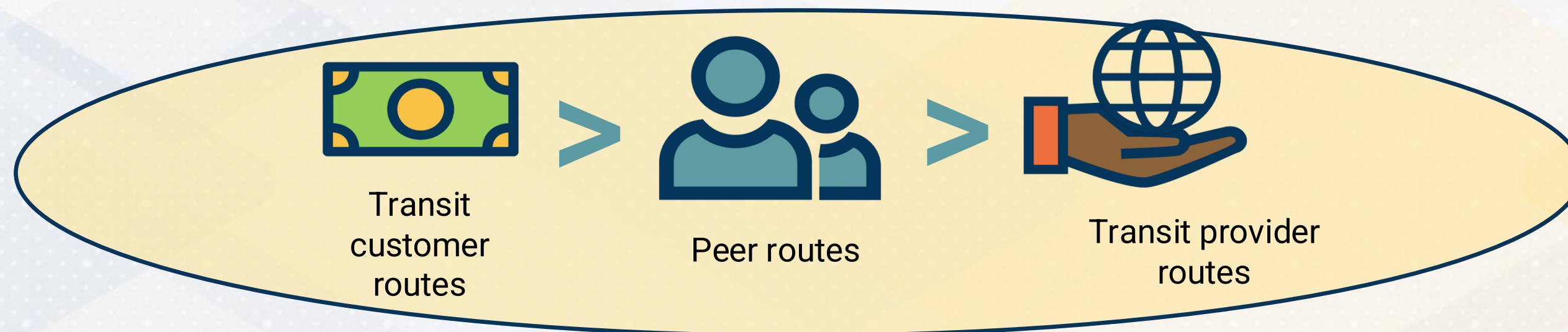
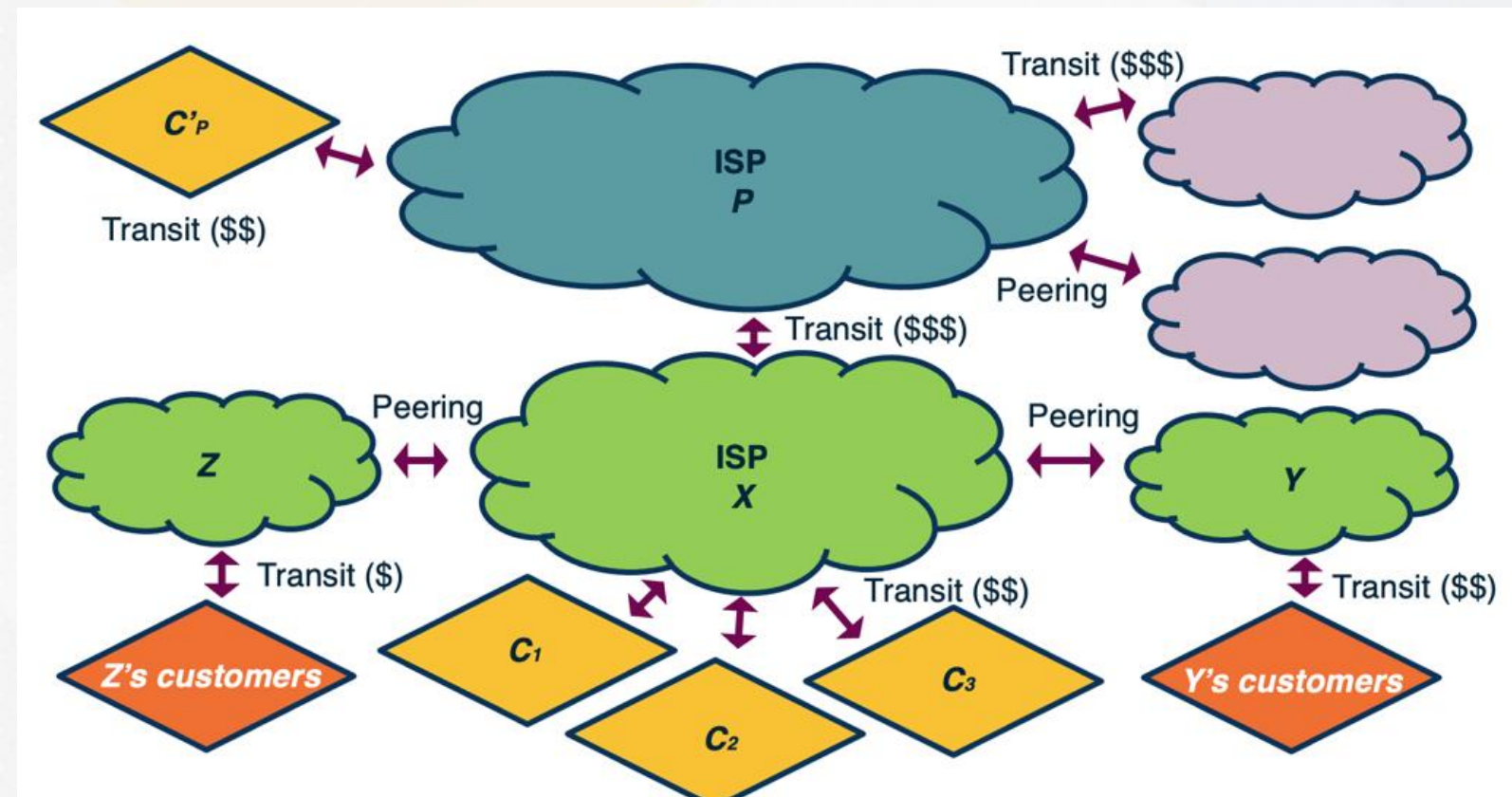
ASes and Internet Interconnection

ASes Businesses Relationships



ASes and Internet Interconnection

BGP Routing Policies, Importing/Exporting Routes



BGP Path Selection Process

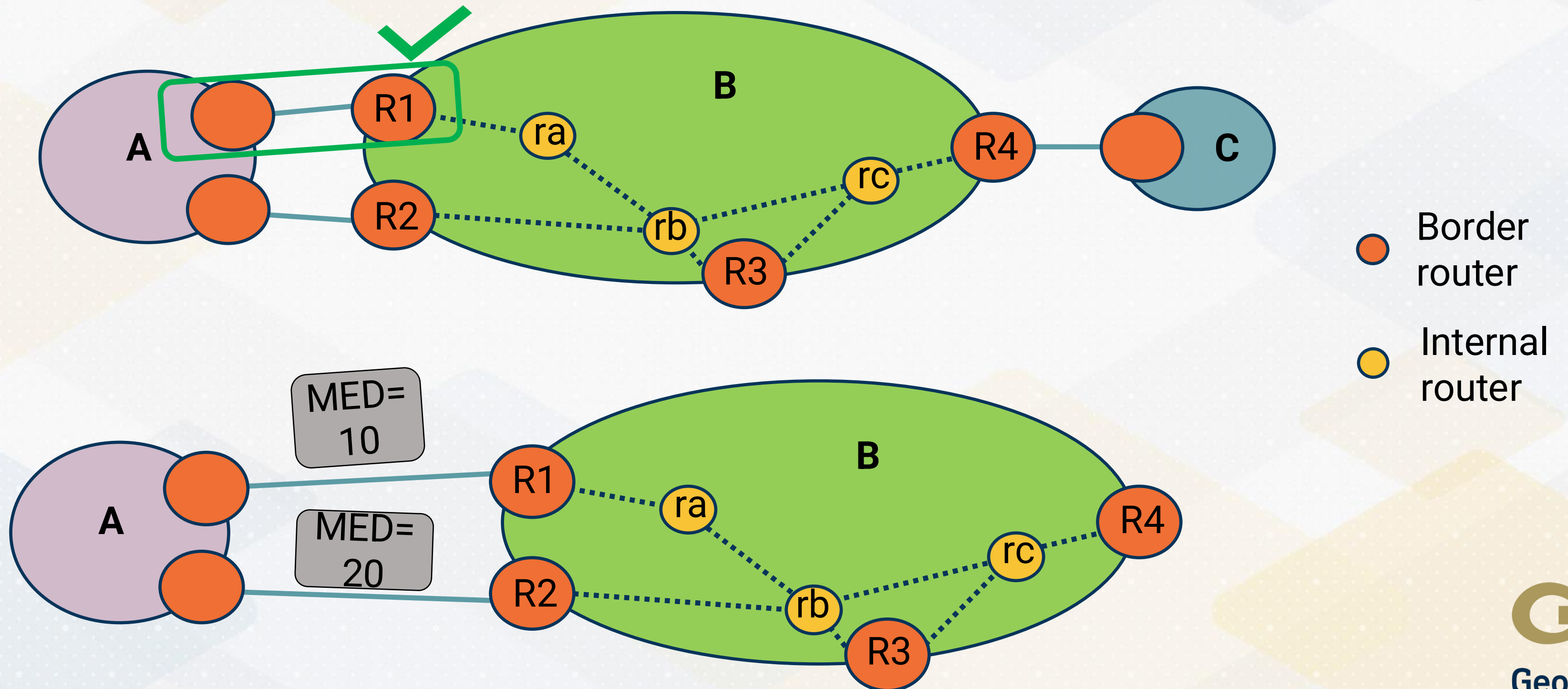
Path Selection and Attributes



Step	Attribute	Controlled by local or neighbor AS?
1.	Highest LocalPref	local
2.	Lowest AS path length	neighbor
3.	Lowest origin type	neither
4.	Lowest MED	neighbor
5.	eBGP-learned over iBGP-learned	neither
6.	Lowest IGP cost to border router	local
7.	Lowest router ID (to break ties)	neither

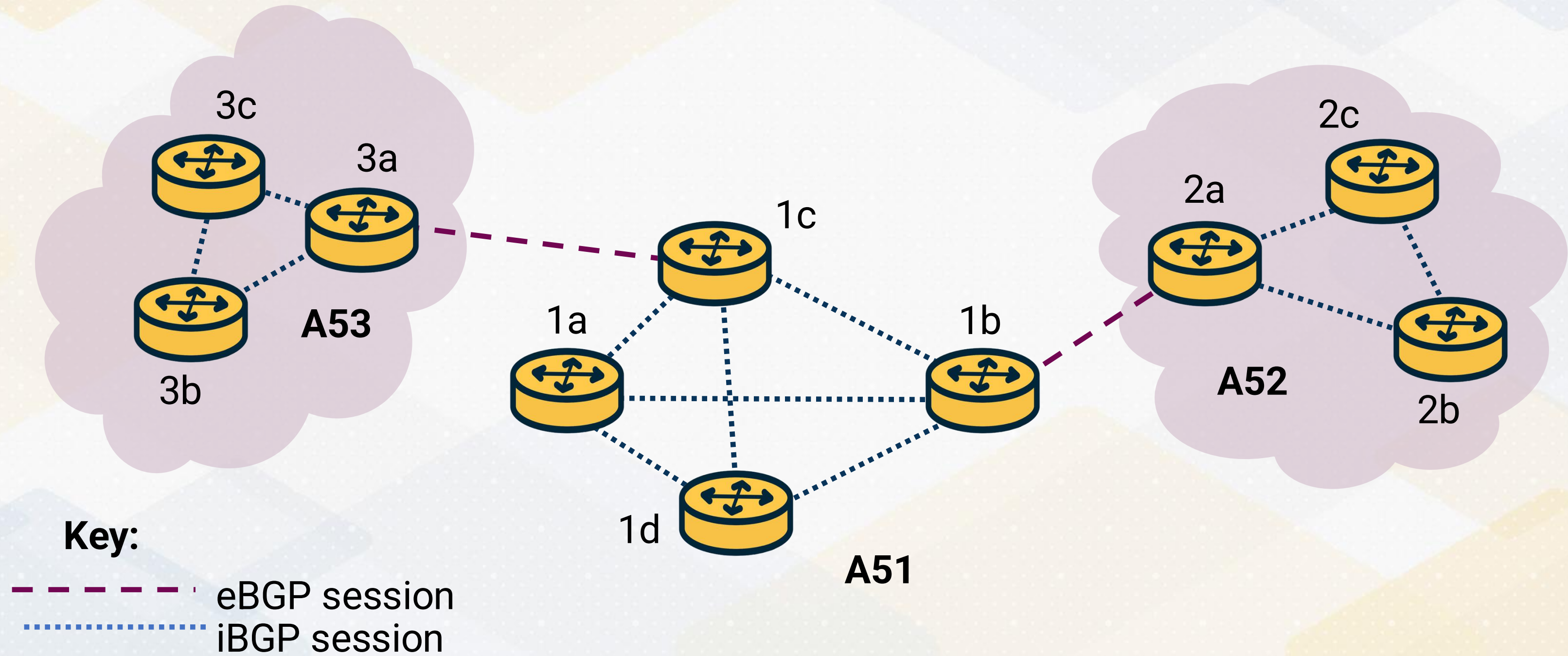
BGP Path Selection Process

Attributes: Local Pref and MED



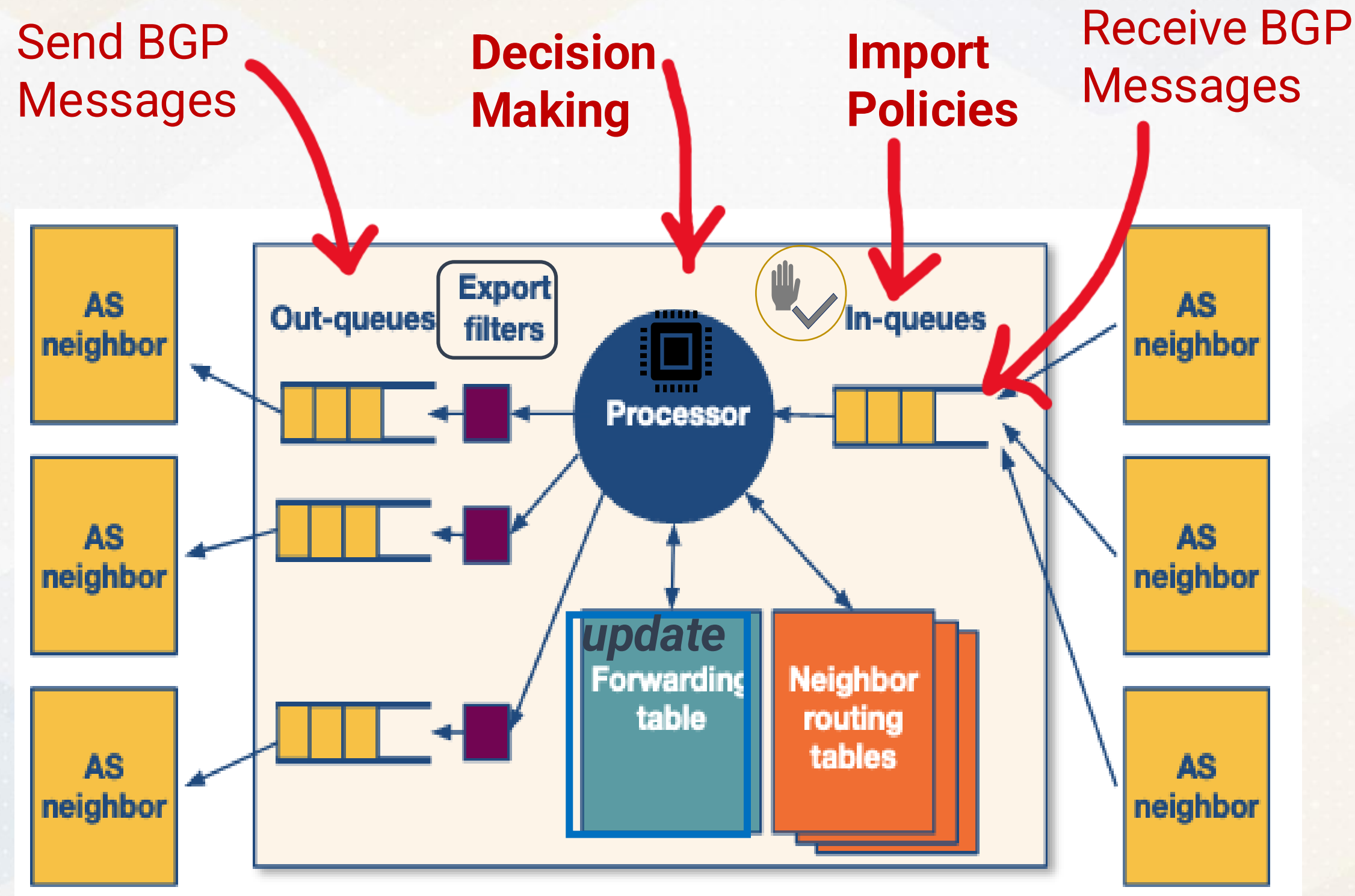
BGP “Flavors” and Mechanics

iBGP, eBGP



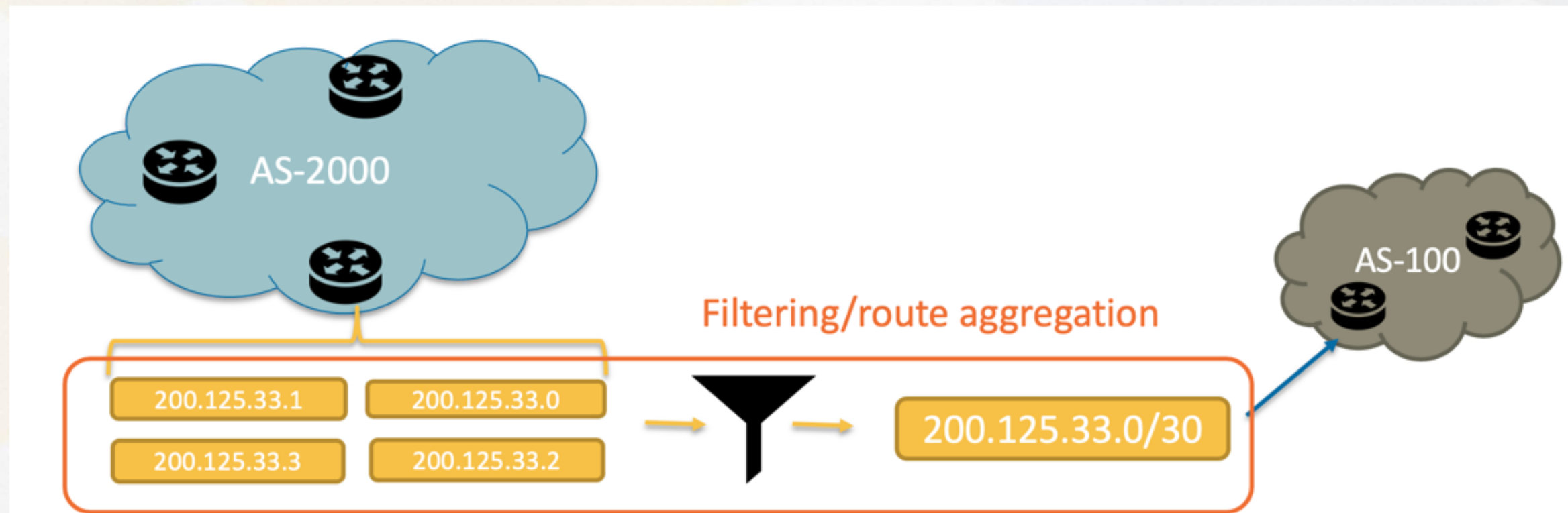
BGP Basics

How A Router Combines Inter and Intra – domain Information



BGP Challenges

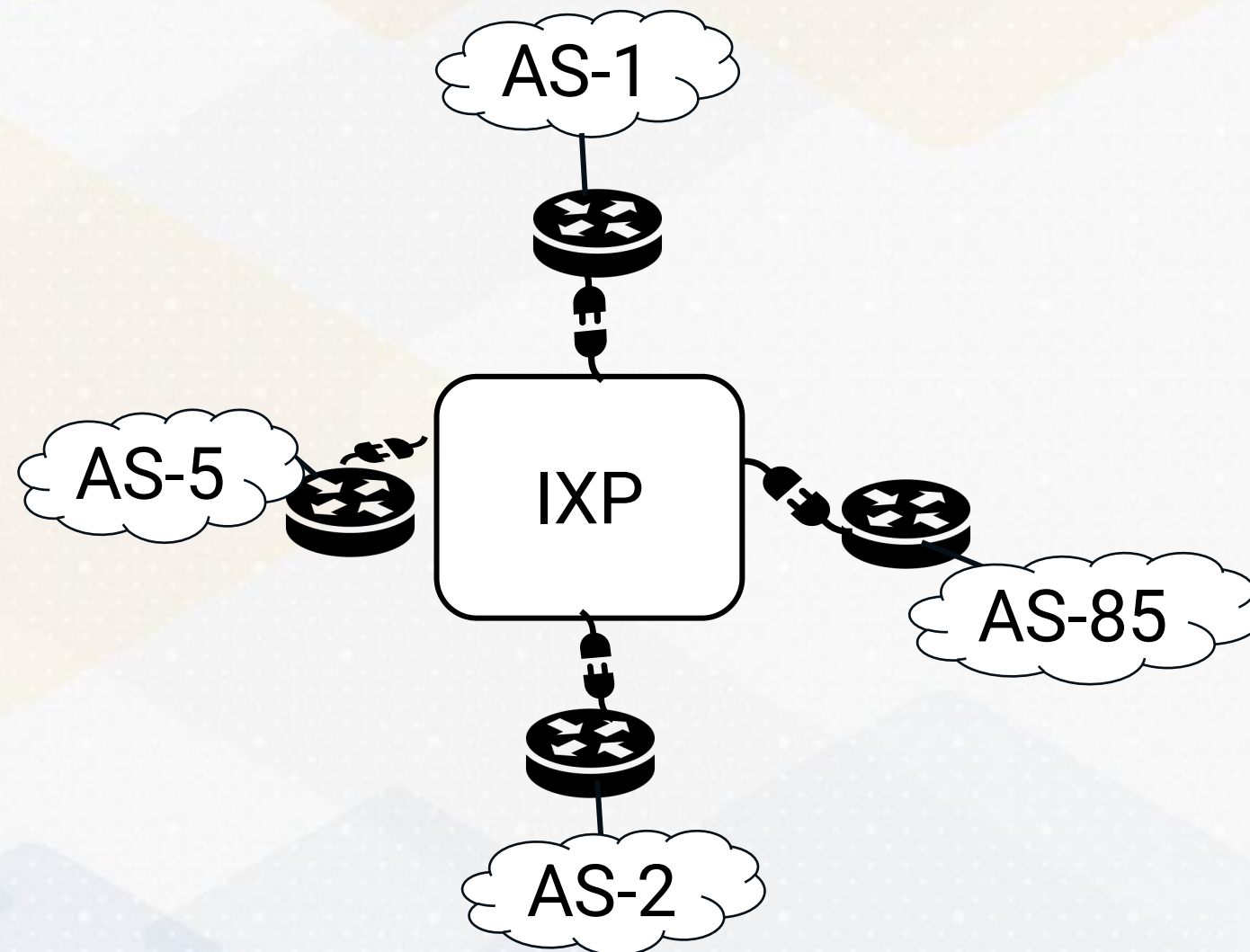
Scalability and Misconfigurations



- **Misconfigs and faults in BGP control-plane:** wrong, excessive, or unstable route announcements/withdrawals
- **Update storms and routing churn:** rapid waves of UPDATE/withdraw messages that propagate across peers
- **Routing table growth pressure:** too many prefixes or overly specific prefixes increase state and complexity
- **Scalability is operational, not automatic:** stability depends on filters, limits, aggregation, and churn controls (with tradeoffs)

Peering at Internet Exchange Points (IXPs)

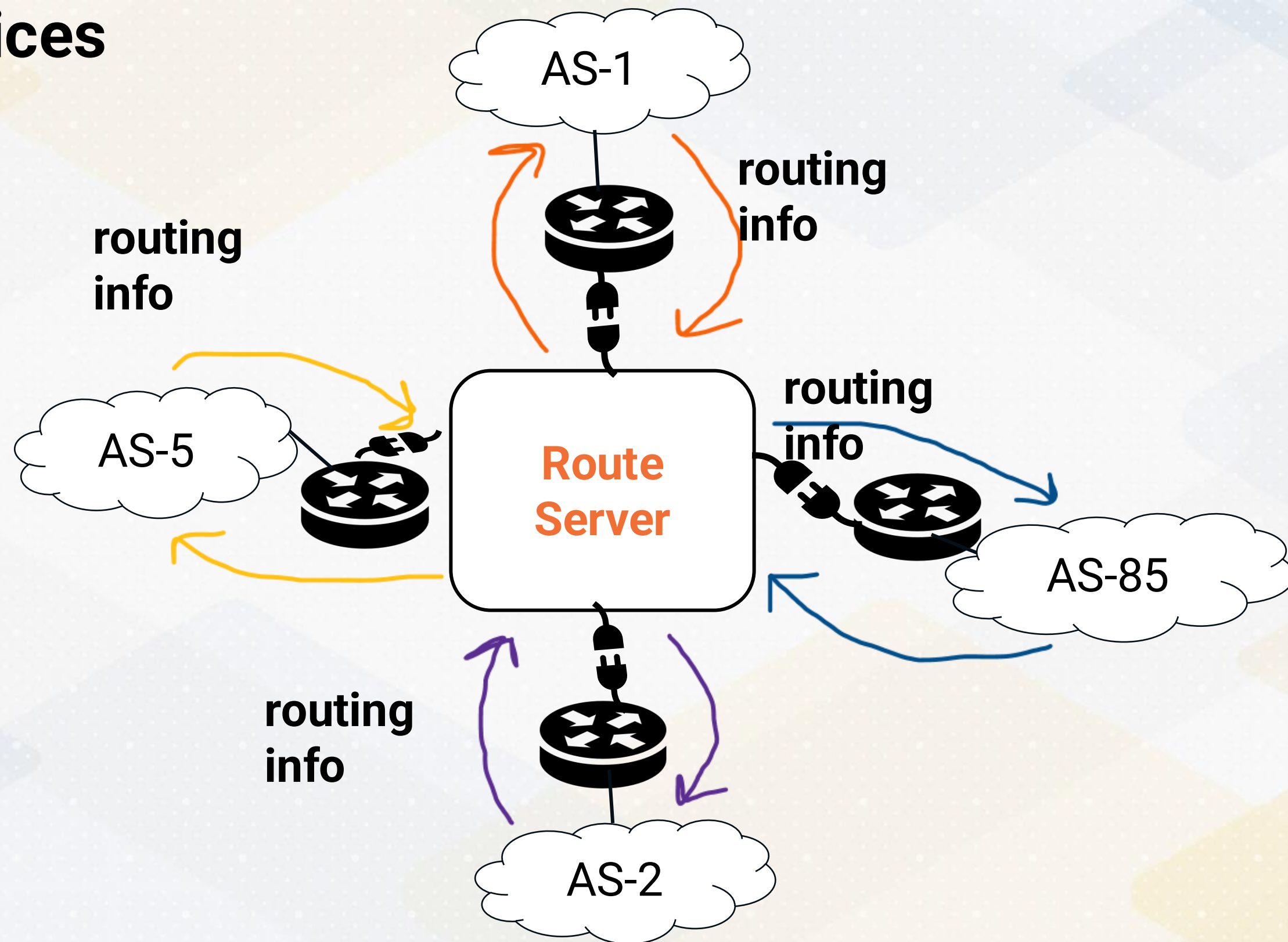
Interconnection at Scale



- **Interconnection infrastructure (IXPs):** physical switching fabric that lets networks (ISPs, CDNs, enterprises) exchange traffic **locally**
- **Participant ASes:** the ASes that connect to the IXP and peer there
- **Peering options:** **public peering** (shared fabric) and **private peering** (direct interconnect)
- **Service offerings:** route server support and **service-level agreements (SLAs)** for managed peering services
- **Remote peering:** access via **resellers** (connect to the IXP without being in the same facility)
- **Security + extras:** **DDoS blackholing** and **free value-added services** (e.g., IRR tools, NTP, DNS/root services, monitoring)

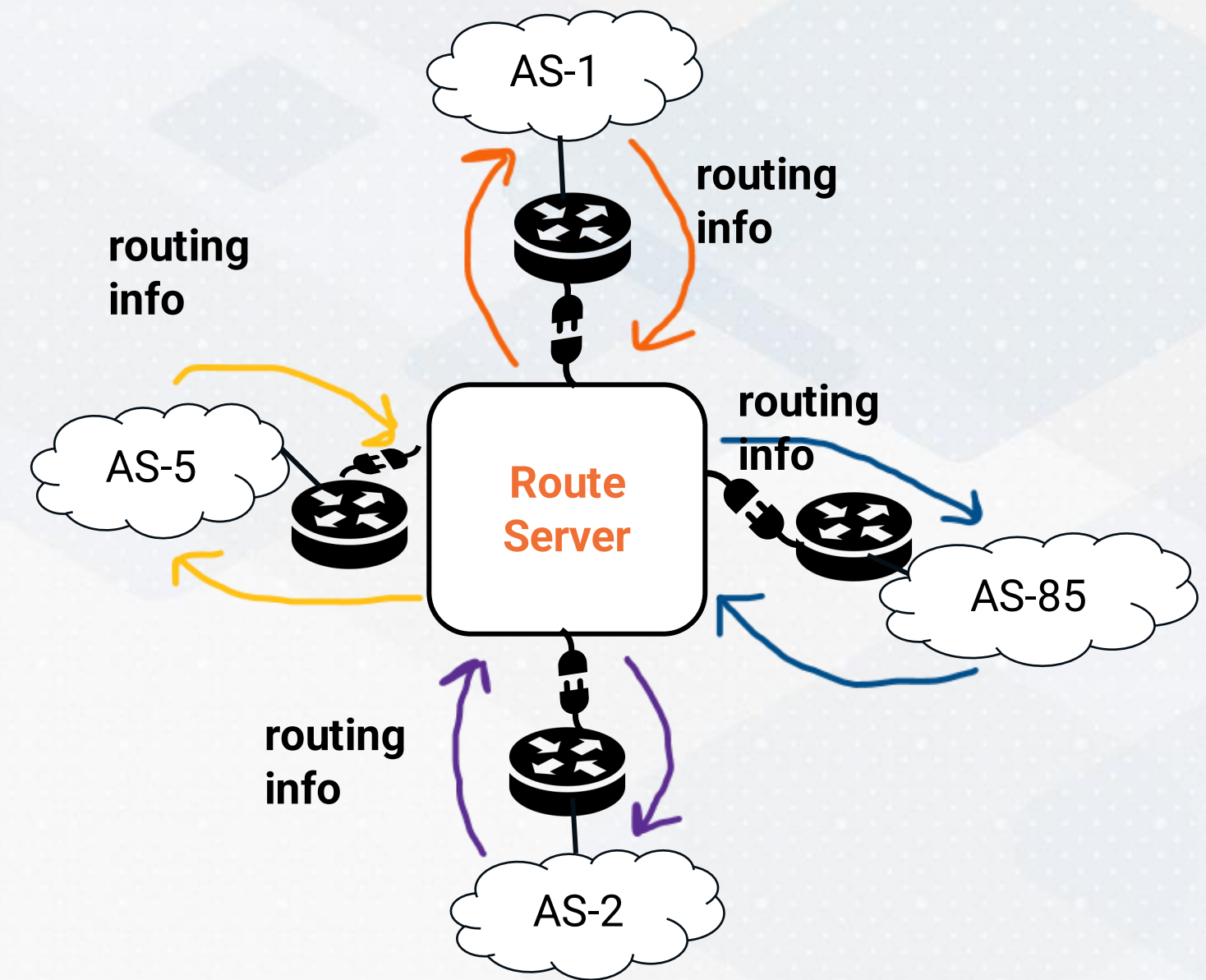
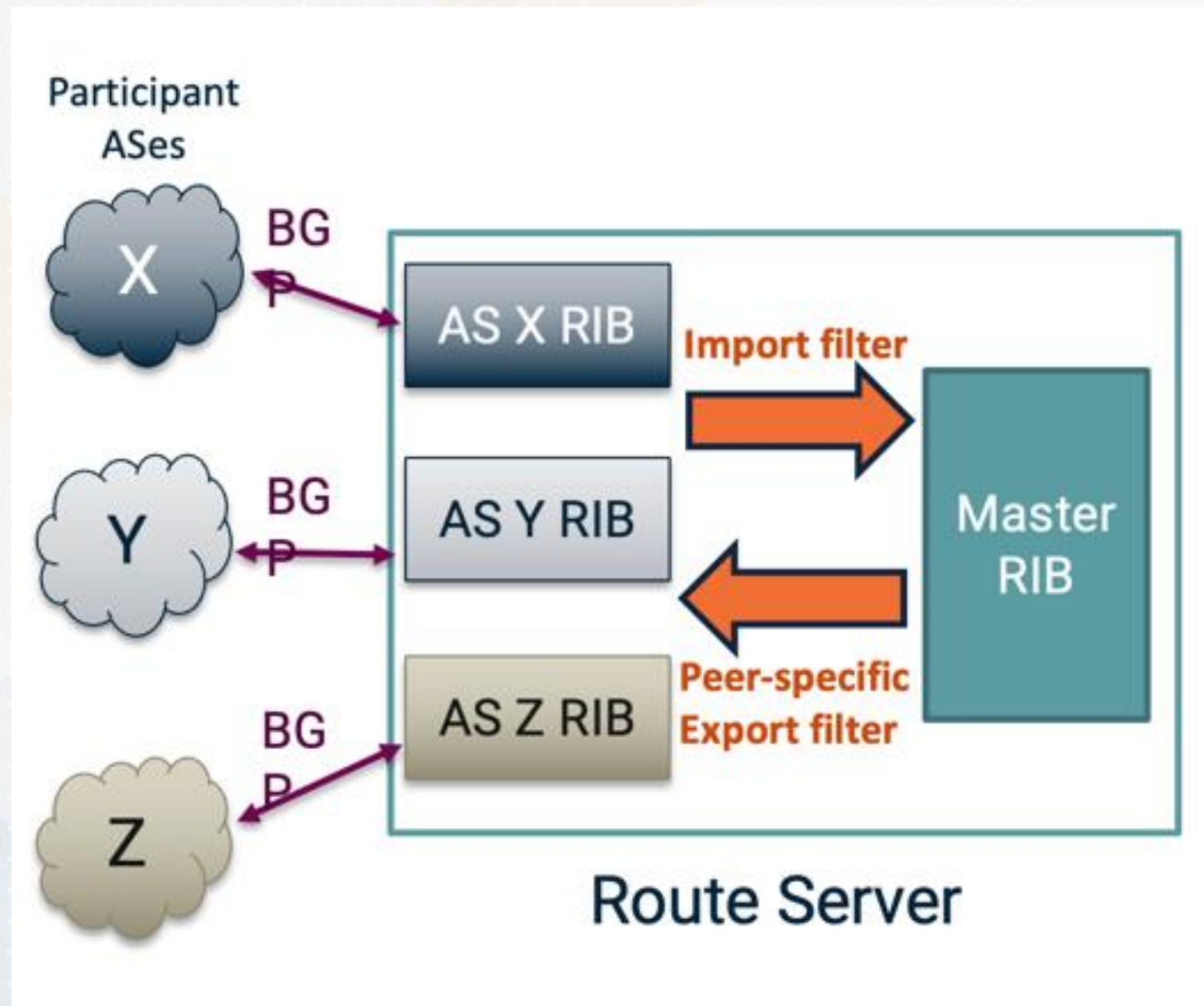
Peering at IXPs

IXPs and Services



Peering at IXPs

How A Route Server Works



Summary

- **Interdomain routing across a “network of networks”:** independently operated Autonomous Systems interconnect, and their economic incentives (cost, revenue, performance) shape which paths are even acceptable—so the AS is the core policy boundary where those choices are enforced.
- **Import and export policies translate business relationships into routing behavior:** networks prefer and propagate routes differently depending on whether they come from customers, peers, or providers, using filters and rankings to avoid “free transit” and align traffic flow with economics.
- **BGP is the control-plane mechanism** that makes this scalable: it uses long-lived TCP sessions to exchange incremental UPDATE/withdraw messages, applies route attributes through a structured decision process, and then installs the selected route into forwarding by combining BGP’s chosen next hop with the IGP’s internal path to build the FIB.
- **Traffic engineering and operations determine how BGP behaves in practice:** LocalPref is the primary knob for controlling outbound exit choice, MED can sometimes influence inbound entry choice, stability relies on safeguards like filters, prefix limits, aggregation, and damping, and IXPs with route servers enable peering to scale without maintaining hundreds of bilateral sessions.